

Cost Behaviour and Terminology

Cost Accounting: Chapter Two

Cost Object

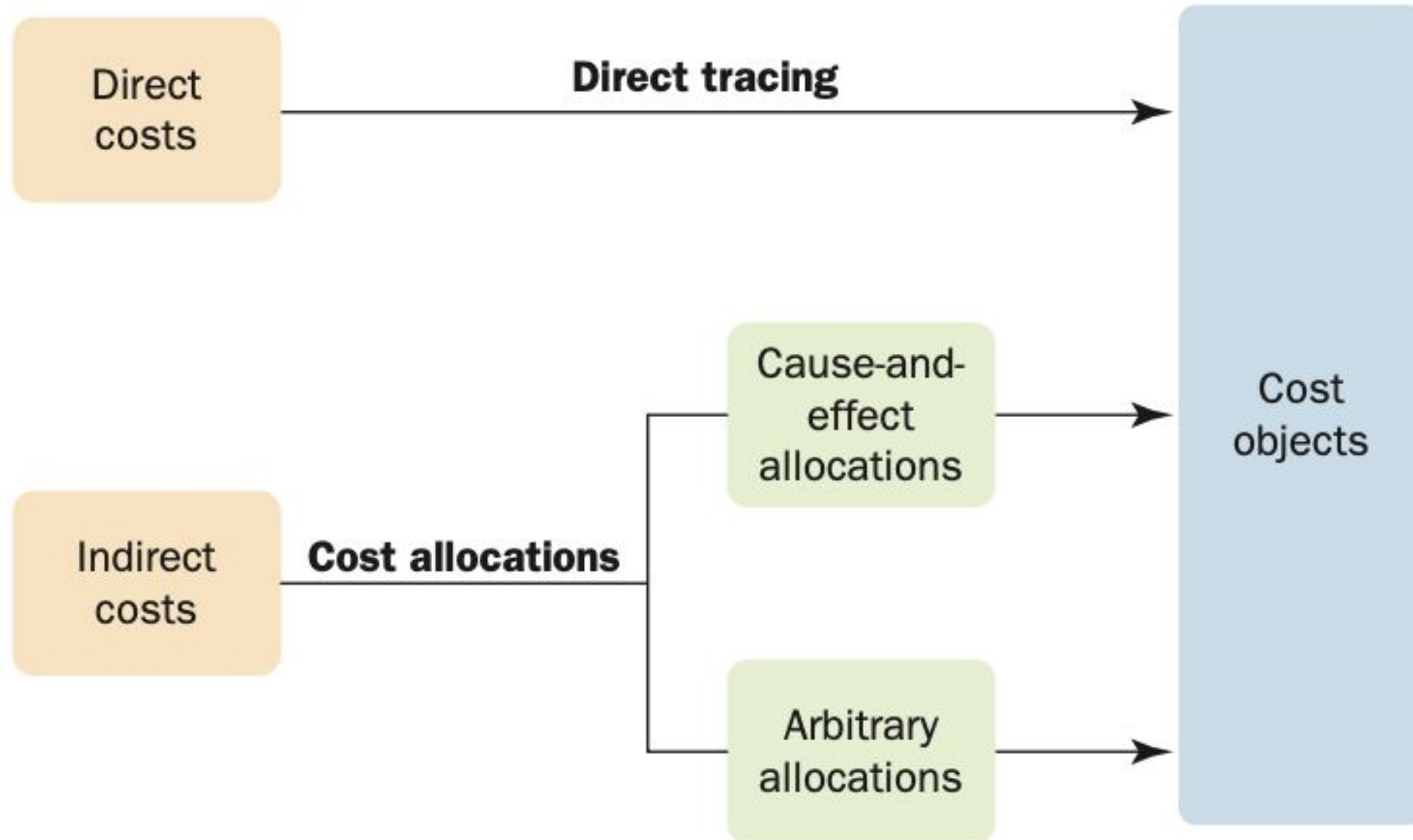
A Cost Object is any item, activity, or entity for which a business wants to separately measure and assign costs.

Industry	Cost Objects
Leather Engineering	Tanned leather , Finished leather, Beam house operations
Leather Product Engineering	Leather bags, Stitching department, A special order of customized leather jackets for a brand
Footwear Engineering	Men's formal shoes, Sole attaching process

Cost Classifications for Assigning Costs to Cost Objects

- **Direct costs** of a cost object are related to the particular cost object and can be traced to it in an economically feasible (cost-effective) way. For example, the cost of Leather, Wages of workers cutting leather, Royalties paid to a designer for a specific branded shoe.
- **Indirect costs** of a cost object are related to the particular cost object but cannot be traced to it in an economically feasible (cost-effective) way. For example, Cleaning supplies for factory floor, Security guards and administrative staff, Factory rent.

Direct Costs and Indirect Costs



Cost Classifications for Manufacturing Companies

- **Direct Materials:** The materials that go into the final product are called raw materials.
- **Direct Labor:** Direct labor consists of labor costs that can be easily (i.e., physically and conveniently) traced to individual units of product.
- **Manufacturing Overhead:** Manufacturing overhead , the third manufacturing cost category, includes all manufacturing costs except direct materials and direct labor. Manufacturing overhead includes items such as indirect materials; indirect labor; maintenance and repairs on production equipment; and heat and light, property taxes, depreciation, and insurance on manufacturing facilities.

Cost Classification s for Manufacturin g Companies

Manufacturing costs

Direct
materials

Direct
labour

Manufacturing
overhead

Prime
cost

Conversion
cost

Non-manufacturing costs

Administrative
overheads

Marketing overhead
(order-getting and
order-filling costs)

Cost Drivers and Relevant Range

Cost Drivers: A cost driver is a variable, such as the level of activity or volume that causally affects costs over a given time span. An activity is an event, task, or unit of work with a specified purpose—for example, designing products, setting up machines, or testing products.

Relevant Range: Relevant range is the band of normal activity level or volume in which there is a specific relationship between the level of activity or volume and the cost in question. For example, a fixed cost is fixed only in relation to a given wide range of total activity or volume (at which the company is expected to operate) and only for a given time span (usually a particular budget period).

Cost Classifications for Predicting Cost Behavior

- A variable cost changes in total in proportion to changes in the related level of total activity or volume.
- A fixed cost remains unchanged in total for a given time period, despite wide changes in the related level of total activity or volume.
- A mixed cost contains both variable and fixed cost elements. Mixed costs are also known as semivariable costs.
- Step Costs: Remain fixed within a range but jump to a new level when capacity changes (e.g., adding a new supervisor when production exceeds a limit).

Methods for Estimating Cost Behaviour of Mixed Cost

- Scattergraph Plot
- The High-Low Method
- The Least-Squares Regression Method

The High-Low Method Exercise

	Patient-Days	Maintenance Cost Incurred
High activity level (June)	8,000	\$9,800
Low activity level (March)	<u>5,000</u>	<u>7,400</u>
Change	<u><u>3,000</u></u>	<u><u>\$2,400</u></u>

The Least-Squares Regression Method

Month	Patient-Days (X)	Maintenance Cost (Y)
Jan	3,000	\$6,200
Feb	4,000	\$6,800
Mar	5,000	\$7,400
Apr	6,000	\$8,100
May	7,000	\$8,900
Jun	8,000	\$9,800

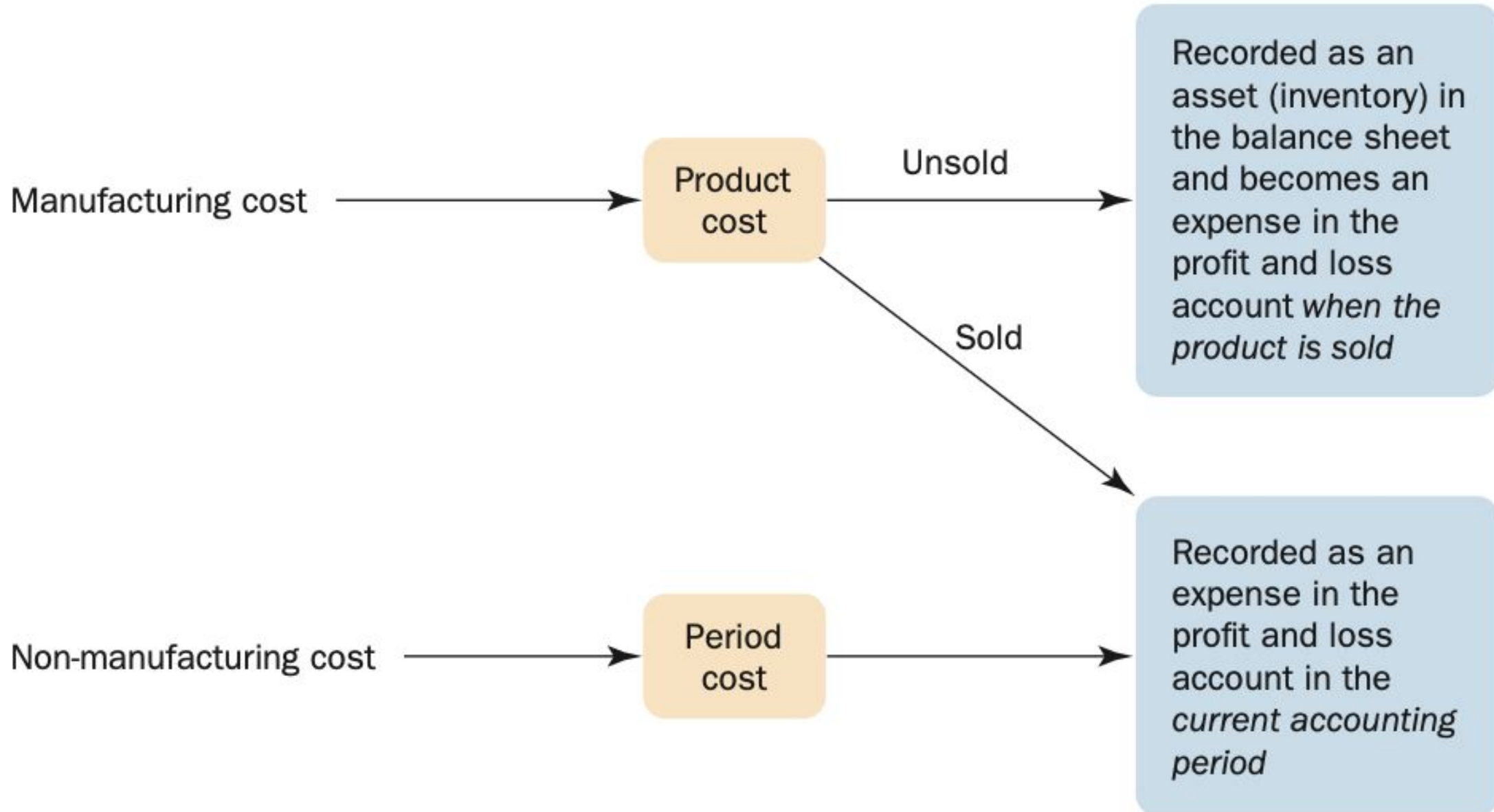
Relationships of Types of Costs

Assignment of Costs to Cost Object	
Cost-Behavior Pattern	Direct Costs
	Indirect Costs
	• Cost object: BMW X5s produced Example: Tires used in assembly of automobile
	• Cost object: BMW X5s produced Example: Power costs at Spartanburg plant. Power usage is metered only to the plant, where multiple products are assembled.
Variable Costs	• Cost object: BMW X5s produced Example: Salary of supervisor on BMW X5 assembly line
	• Cost object: BMW X5s produced Example: Annual lease costs at Spartanburg plant. Lease is for whole plant, where multiple products are produced.
Fixed Costs	

Types of Inventory

1. Direct materials inventory. Direct materials in stock and awaiting use in the manufacturing process (for example, computer chips and components needed to manufacture cellular phones).
2. Work-in-process inventory. Goods partially worked on but not yet completed (for example, cellular phones at various stages of completion in the manufacturing process). This is also called work in progress.
3. Finished goods inventory. Goods (for example, cellular phones) completed but not yet sold.

Period and Product Costs



Economic Cost Patterns

- **Explicit Costs:** Actual monetary payments (e.g., wages, raw material, rent).
- **Implicit Costs (Opportunity Costs):** The value of benefits foregone when choosing one alternative over another (e.g., using own building for factory instead of renting it out).
- **Incremental Costs:** Additional costs incurred if a particular action is taken (e.g., cost of producing an extra batch).
- **Sunk Costs:** Past costs already incurred, cannot be recovered, and should not affect future decisions (e.g., old machine purchase price).
- **Marginal Cost:** Cost of producing one additional unit.
- **Differential Cost:** Difference in total cost between two alternatives.

Accounting Cost Patterns

- **Historical Costs:** Actual costs incurred and recorded (e.g., wages paid last month).
- **Standard Costs:** Predetermined/estimated costs used for cost control (variance analysis).
- **Book Costs:** Costs recorded in books but not paid out in cash during the period (e.g., depreciation).
- **Out-of-pocket Costs:** Actual cash outflows for expenses (e.g., salaries, utilities).
- **Controllable & Uncontrollable Costs:** Costs managers can influence (e.g., materials usage) vs. those they cannot (e.g., factory rent fixed by lease).

Why Knowledge of Cost Behavior is Needed in Product Costing:

1. **Accurate Cost Allocation:** Helps separate fixed and variable portions to properly assign costs to products.
2. **Break-even & Profit Planning:** Understanding how costs behave supports CVP (Cost-Volume-Profit) analysis and break-even calculations.
3. **Decision Making:**
 1. *Make or Buy Decision:* Knowing which costs are variable vs. fixed helps managers decide whether to outsource.
 2. *Pricing:* Contribution margin ($\text{Sales} - \text{Variable Costs}$) analysis depends on cost behavior.
 3. *Product Mix:* Helps identify which product yields higher contribution per limiting factor.
4. **Budgeting and Forecasting:** Cost projections depend on how costs will behave with changes in activity.
5. **Controlling Costs:** Managers can target controllable variable costs and monitor fixed cost commitments.